

Team name:	Optimize
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Hardware

Name of the Local Currency	Total Cost Local Currency	Total Cost (\$) U.S.A. Dollars
New Israeli Shekel NIS	₪3,417.00	\$1,195.21

#	Component	Part name	Autor	Source	Quantity	Status	Custom bulit/KIT/Bought	Unit cost (Local Currency)	Unit cost (US Dollars)	Total Cost (Local Currency)	Total Cost (USA Dollars)
1	Arduino	Arduino mega 2560 board	Arduino	hackstore	1	Reused	Bought	₪175.00	\$59.91	₪77.00	\$59.91
2	Raspberry pi 4	Raspberry pi 4, 8 GB	Raspberry Pi Ltd	piitel	1	New	Bought	₪750.00	\$256.78	₪750.00	\$256.78
3	Stepper motor	Stepper motor 28BYJ-48	Maintex	hackstore	1	Reused	Bought	₪30.00	\$10.27	₪30.00	\$10.27
4	Stepper motor driver	ULN2003	Maintex	hackstore	1	Reused	Bought	₪25.00	\$8.56	₪25.00	\$8.56
5	Camera	USB camara 2MP	WaveShare	4project	2	New	Bought	₪178.00	\$60.94	₪356.00	\$121.88
6	LCD screen	Lcd 1602 with I2C module, Blue Light	Hitachi	hackstore	1	Reused	Bought	₪20.00	\$6.85	₪20.00	\$6.85
7	Distance sensor	GY53-VL53L0X	STMicroelectronics	see-sys	4	New	Bought	₪58.00	\$19.86	₪232.00	\$79.43
8	Gyroscope	Gyroscope BNO055	Bosch Sensortec	aliexpress	1	New	Bought	₪24.00	\$8.22	₪24.00	\$8.22
9	DC motors	DC motors 25ga-370 242rpm 4mm	Generic Spec	hackstore	4	New	Bought	₪80.00	\$27.39	₪320.00	\$109.56
10	Omni wheels	70mm Omni Wheel - 6mm Bore	GTF Robots	robotshop	4	Reused	Bought	₪186.00	\$63.68	₪744.00	\$254.72
11	LEDs	Neo pixel RGB 5V	Adafruit Industries	hackstore	9	New	Bought	₪15.00	\$27.39	₪135.00	\$46.22
12	TCS34725	TCS34725 RGB Color Sensor Module	Adafruit Industries	see-sys	1	New	Bought	₪34.00	\$5.14	₪34.00	\$3.42
13	Voltage regulator	Custom made	unknown (is used in the school for years)	pcbway	1	Reused	Bought	₪10.00	\$3.42	₪10.00	\$3.42
14	Voltage card	Custom made	unknown (is used in the school for years)	pcbway	1	Reused	Bought	₪10.00	\$3.42	₪10.00	\$3.42
15	Cable	Cable USB type A to type B	Generic	hackstore	1	Reused	Bought	₪15.00	\$5.14	₪15.00	\$5.14
16	Motors driver	Motor driver L298N	STMicroelectronics	hackstore	2	Reused	Bought	₪35.00	\$11.98	₪70.00	\$23.97
17	Cables	Various cables		Local electronics store	1	New	Bought	₪100.00	\$34.24	₪100.00	\$34.24
18	Battery	Lithium Ion 12v 2000mA	belline	belshop	1	Reused	Bought	₪175.00	\$59.91	₪175.00	\$59.91
19	Portable Battery	AVA+ 10000mAh JP399	AVA+	misradia	1	Reused	Bought	₪40.00	\$13.69	₪40.00	\$13.69
20	Connecting parts	Screws, nuts, washers, ...		Local tool store	1	Reused	Bought	₪150.00	\$51.36	₪150.00	\$51.36
21	Laser cut parts	platets, battry holder	custom made	Laser cutter	1	New	Custom bulit	₪40.00	\$13.69	₪40.00	\$13.69
22	3D printed parts	Wheels, connectors, holders, kits & holder	custom made	3d printer	1	New	Custom bulit	₪60.00	\$20.54	₪60.00	\$20.54

Software

#	Name	Software's Tool/Library	Description	Source	Author	Cost (Local Currency)	Cost (USA Dollars)
1	Tinckercad	Software's Tool	Online platform for simulating circuits	Tinkercad	Autodesk	FREE	FREE
2	Fusion 360	Software's Tool	3D CAD shell design	Autodesk Fusion 3D CAD ,	Autodesk	EDUCATION	EDUCATION
3	Git	Software's Tool	Version control for managing functions	Git - Descargas	Software Freedom	FREE	FREE
4	Visual Studio Code	Software's Tool	General purpose text editor	Visual Studio Code	Microsoft	FREE	FREE
5	Arduino IDE	Software's Tool	IDE used for writing, compiling and uploading arduino code	Arduino IDE	Arduino	FREE	FREE
6	Wire.h	Library	Built-in library for I2C communication between the microcontroller and peripheral sensors.	Wire	Nicholas Zambetti	FREE	FREE
7	Adafruit_Sensor.h	Library	Unified base library providing a standard interface for various Adafruit sensors.	adafruit_sensor	Adafruit Industries	FREE	FREE
8	Adafruit_BNO055.h	Library	Dedicated library for reading orientation, acceleration, and gyro data from the BNO055 sensor.	adafruit_BNO055	Adafruit Industries	FREE	FREE
9	Adafruit_TCS34725.h	Library	Library for interfacing with the TCS34725 color sensor to read RGB and light data.	adafruit_tcs34725	Adafruit Industries	FREE	FREE
10	Stepper.h	Library	Built-in library for controlling standard unipolar or bipolar stepper motors.	stepper	Ryan Orendorff	FREE	FREE
11	Adafruit_NeoPixel.h	Library	Library for controlling individually addressable RGB LED strips and rings (NeoPixels).	adafruit_neopixel	Phillip Burgess	FREE	FREE
12	LiquidCrystal_I2C.h	Library	Library for managing character LCD screens connected via an I2C backpack module.	LiquidCrystal	Frank de Brabander	FREE	FREE
13	Numpy	Library	Array and math library	NumPy	Numpy developers	FREE	FREE
14	OpenCV	Library	Image proccessing and computer vision algorithms	OpenCV	OpenCV developers	FREE	FREE
15	pyserial	library	Serial communication using python	pyserial	Chris Liechti	FREE	FREE
16	Flask	library	Web application framework	Flask	Pallets	FREE	FREE